

Total Customized Alloplastic Reconstruction for Treatment of Severe Temporomandibular Joint Pathologic Conditions: A Case Series of Combined Intraoral and Extraoral Approach.

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Temporomandibular joint (TMJ) reconstruction with customized alloplastic implants has become a safe and effective treatment option of TMJ end-stage pathology with excellent outcomes reported in the literature. The purpose of this study is to report 5 cases of severe TMJ pathology and customized alloplastic reconstruction using a combined intraoral approach and extraoral approach. Four patients with TMJ involved for benign tumor and one patient with severe TMJ resorption were enrolled. Compromised joints were replaced with customized prosthesis under general anesthesia using an association of intraoral approach/extraoral approach. An implant handpiece with adapted drills for bone drilling and the insertion of screws was used to fixate the mandibular component intraorally; the fossa component was inserted via preauricular approach. The hemimandibulectomies/codilectomy with safety margin were successfully performed and for 2 patients Orthognathic Surgery was also required. Follow-up period was from 15 to 28 months (average 22 months), with no history of surgical site infection or damage to the prostheses. Occlusal relationship and function, as well as facial symmetry were kept stable in all patients. The combination of an intraoral and extraoral approach for total TMJ replacement with customized prosthesis may be an alternative and reliable strategy for pathologic reconstruction, keeping function and reducing aesthetic damage.

Concomitant Temporomandibular Joint Replacement and Orthognathic Surgery.

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The treatment of patients with severe temporomandibular joint (TMJ) disorders and associated skeletal discrepancies presents a complex challenge for oral and maxillofacial surgeons. It is widely recognized that TMDs can impact the stability and outcomes of surgical treatments for dentofacial deformities. Consequently, addressing TMDs prior to or in conjunction with orthodontic or surgical interventions may be necessary to achieve optimal and long-lasting results. Alloplastic temporomandibular joint replacement (TMJR) and orthognathic surgery have emerged as the standard approach due to their predictability, long-term stability and excellent outcomes when addressing end-stage TMJ disease in conjunction with DFDs as it provides a comprehensive solution to address both functional and aesthetic aspects of these patients' conditions. Understanding the appropriate utilization of TMJR in conjunction with orthognathic surgery can lead to improved treatment planning and successful outcomes for patients with complex TMJ disorders and associated dentofacial deformities. This review aims to discuss the indications, preoperative evaluation, staging, sequencing, and surgical considerations involved in utilizing alloplastic TMJ replacement in the presence of dentofacial deformities.

Temporomandibular joint reconstruction with alloplastic prosthesis: the outcomes of four cases.

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The purpose of this study is to evaluate the outcomes of four patients receiving stock Biomet TMJ

prosthesis for reconstruction of the TMJs. TMJ reconstruction with stock Biomet TMJ prosthesis was performed in four patients who had joint damages by trauma, tumor, resorption, and ankylosis, which represent the indications of alloplastic prosthesis. Loss of condyle from trauma and resorption of joint are good indications for prosthesis, but the patients should be informed about limitation of jaw movement. In case of structural damage of TMJ by tumor, tumor recurrence should be considered before planning TMJ reconstruction. Considering heterotopic bone formation in case of ankylosis, periodic follow-up and special surgical technique are required. Given careful treatment planning and understanding the functional limitation of TMJ prosthesis, alloplastic prosthesis is a safe and effective management option for the reconstruction of TMJs.

A new three-dimensional, print-on-demand temporomandibular prosthetic total joint replacement system: Preliminary outcomes.

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The aim of this study is to present the preliminary clinical data on the OMX Temporomandibular Joint (TMJ) Prosthetic total joint replacement system. A prospective, cohort, clinical study was undertaken of consecutive adult patients with Category 5 end-stage joint disease who were implanted with the OMX TMJ prosthesis between May 2015 and April 2017. A total of 50 devices were implanted in 38 patients, with 12 patients receiving bilateral prosthetic joints. There were 31 females and 7 males in this cohort, who ranged in age from 20 to 66 years, with a mean of 43.8 years (± 14.0 years). Ten of the 50 prosthetic joints (20%) were fully customized, while the remaining were patient matched using virtual planning software. Based on a mean follow-up period of 15.3 months (range 12-24 months) following the TMJ total joint replacement, preliminary results suggest the OMX TMJ prosthesis has made a positive impact on clinical outcomes, with a mean 74.4% reduction in joint pain levels and significant improvements ($p < 0.05$) in jaw function as measured by the visual analogue scales for mouth opening (30.8%), diet (77.1%), and function (59.2%). No device failures were reported during the study period. This study suggests that the print-on-demand OMX TMJ prosthesis, designed for rapid delivery of both patient-matched and fully customize devices, represents a safe, reliable and versatile implantable joint replacement system for the treatment of category 5 end-stage TMJ disease.

Alloplastic Temporomandibular Joint Reconstruction Following Recurrent Ameloblastoma Resection.

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Ameloblastomas are benign tumors that most commonly affecting the mandible. The current standard of treatment for ameloblastomas is resection followed by reconstruction that has historically been accomplished through the use of a microsurgical vascularized flaps taken from the iliac crest or fibula. Alloplastic reconstruction methods have gained popularity over recent years with success reported in the reconstruction of many pathologies, including ankylosis, condylar fracture, neoplasia involving extensive resection, severe inflammatory/degenerative temporomandibular joint (TMJ) disease, and congenital TMJ abnormalities. The authors present a patient who successfully underwent ameloblastoma resection and TMJ reconstruction with a custom TMJ Concepts alloplastic implant. The authors also present a review of the literature on alloplastic TMJ reconstruction following ameloblastoma resection. To our knowledge, this

is the second report in the literature on the use of a TMJ Concepts implant after ameloblastoma resection.

Bilateral alloplastic prostheses for temporomandibular joint reconstruction in a patient with ankylosing spondylitis.

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Ankylosing spondylitis (AS) or Bechterew disease is a chronic, usually progressive, systemic inflammatory joint disease, which predominantly affects the spine and sacroiliac joints. In these joints, early inflammatory changes are followed by lumbosacral pain and progressive restriction of spinal movement associated with radiologically visible intervertebral ossification. Peripheral joint involvement occurs in 10 to 30% of patients and shows a predilection for the shoulders, knees, ankles, feet, and wrists. Temporomandibular joint (TMJ) involvement has been described, and its reported frequency varies from 11 to 35%. However, ankylosis is uncommon with a single documented case utilizing an alloplastic prosthesis for total joint replacement. A case report of bilateral ankylosis of the jaw treated with alloplastic prostheses for total TMJ replacement using a Brazilian system in a patient with AS is presented.

[A primary application and evaluation of temporomandibular joint replacement with stock prosthesis].

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To evaluate the effect of total joint replacement in treatment of temporomandibular joint(TMJ) osteoarthropathy with stock prostheses. Six female patients involving 10 joints (2 unilateral and 4 bilateral), with an average age of 59 years old, were involved in this study. Three patients (5 joints) were diagnosed as internal derangement in V stage depending on MRI, 3D-CT findings and clinical characteristics. The other 3 patients (5 joints) had histories of failed temporomandibular joint operation using costochondral graft or temporalis fascial flap. The maximal mouth opening was 1.9 cm on average (range, 1.0 to 2.9cm). All the joints were replaced with Biomet standard prosthesis under general anesthesia. The follow-up period was from 7 to 49 months (average, 17.5 months). All the operations were successfully performed. Heterotopic ossification happened in a bilateral case 1 year postoperatively. One patient with bilateral joint disease complained of severe uncomfortable feeling in the region of the ears and the temples, although there was no significant positive signs according to an ENT examination. Pain relief of the joint and mouth opening improvement were significant in 4 patients. No failure was noted secondary to infection or loosening of the prostheses. The occlusal relationship kept stable postoperatively in all cases. Total TMJ joint replacement with standard prosthesis is a good choice for TMJ reconstruction. It can significantly reduce joint pain and the mouth opening limitation resulted from osteoarthritis. Long-term result remains to be evaluated based on a long-term follow-up.

Alloplastic temporomandibular joint replacement: present status and future perspectives of the elements of embodiment.

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Medical device embodiment involves the following elements: materials, design, and manufacturing. Failure of any one of these elements can result in failure of the device, despite the others being satisfactory. The abundance of clinical and basic science literature published since 1986, demonstrates the safety and efficacy of alloplastic temporomandibular joint replacement (TMJR). Currently, there are 19 countries producing 41 TMJR devices. More than 75% are custom designed, and 27% are additively manufactured. In light of the increasing number of TMJR devices being designed and manufactured around the world, this paper will discuss TMJR embodiment so that clinicians understand their present status as well as the prospects for the future of new and/or improved TMJR devices, to ensure that these devices continue to be safe and effective long-term surgical options for the management of end-stage TMJ pathologies.

Evaluation of temporomandibular joint total replacement with alloplastic prosthesis: observational study of 27 patients.

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Background Temporomandibular joint (TMJ) total replacement with alloplastic prostheses has been performed since 1960s. Research in these last two decades has achieved important improvement in the development of biomaterials, design, adaptation, and fixation of the prosthesis components. **Objective** To evaluate total TMJ replacement with alloplastic prostheses. **Methods** We studied 27 patients, between 19 and 73 years old, who had total TMJ uni- or bilateral replacement surgery with custom-made alloplastic prostheses manufactured by TMJ Concepts, Inc. (Ventura, CA) between 1996 and 2011. The general data and preoperative measurements were taken from medical records. Subjective data related to pain, diet consistency, and current quality of life were collected with a questionnaire answered by the patients; measures of maximum interincisal opening and lateral mandibular movements were obtained from direct examination. **Results** We found significant difference ($p < 0.05$) comparing pre- to postoperative results about improvement in mouth opening, pain relief, and satisfaction with the surgery and diet consistency. **Conclusion** The results of this study show that total alloplastic TMJ prostheses are an efficient, safe, and stable long-term solution for patients who need TMJ total replacement.

Total alloplastic temporomandibular joint replacement: the Czech-Slovak initial experience.

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In 2004, total alloplastic temporomandibular joint (TMJ) replacement began in the Czech Republic and Slovakia. This paper presents initial subjective and objective data compiled between 2005 and 2009 from those cases. Data were collected from 27 patients (38 joints) reconstructed with the Biomet-Lorenz stock and custom TMJ prostheses during a mean follow-up period of 24 months. The variables of pain and mouth opening were evaluated pre- and postoperatively. Patients classified pain on a scale of 0-5 (none - unbearable). The extent of opening was investigated by a physician (the distance between the points of the incisors on the upper and lower jaw was measured). The most common indication for replacement was ankylosis. There was an improvement in pain score in 15 patients. 4 patients reported worsening of pain and 8 patients did not complain of pre- or postoperative pain. Mandibular opening increased from a

mean of 17.7 mm preoperatively to a mean of 29.1mm postoperatively. There were complications related to the surgery, but no significant complications related to the devices. Total alloplastic TMJ replacement appears to be a safe and effective method of reconstruction in the patients in this initial study.
